# Mattia Volpato

Software Engineer | **Data Engineering • Algorithms & Data Structures • Distributed Systems**

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## Summary

Software Engineer with experience in distributed systems, data-intensive applications and cloud-native software. Currently working as a Data Engineer at Quantyca, Data at Core, developing enterprise-scale data platforms while maintaining a strong focus on software engineering, algorithms and system design. Interested in advanced algorithms and data structures, distributed systems and large-scale software infrastructures.

## Education

**University of Milano Bicocca**, Master of Science in Computer Science – Milan, Italy — Oct 2023 – Sept 2025
Final Grade 110/110 cum laude

- Thesis: Delay Prediction in Supply Chains
- A Hybrid Graph-Based and **Machine Learning** Approach. Developed at Cefriel within the M4ESTRO Horizon Europe project.

**University of Milano Bicocca**, Bachelor of Science in Computer Science – Milan, Italy — Sept 2020 – July 2023
Final Grade 110/110 cum laude

- Thesis: Wavefront algorithm for sequence alignment on graph structures. Developed a **Rust** prototype for efficient graph-based sequence analysis.

## Experience

**Data Engineer**, Quantyca | Data at Core – Monza, Italy — Oct 2025 – present, 10 months

Data Engineer designing scalable software and data-intensive platforms for enterprise customers, combining backend engineering, distributed computing and modern data engineering practices.

- Developed a scalable alerting platform for Prelios on **Databricks**, designing a YAML-to-Jinja framework with schema validation and type checking to generate maintainable alert definitions.
- Implemented the distributed execution engine, executing parameterized **PySpark**/**SQL** workloads safely and publishing alerts through Azure Service Bus while optimizing distributed query performance.
- Designed and maintained enterprise data products for TeamSystem, covering the complete lifecycle from ingestion and transformation to analytics within a **Data Mesh** architecture.
- Built scalable ETL pipelines and analytical workloads using **Databricks**, **Apache Spark**, **PySpark** and **SQL**, improving data quality, governance and self-service analytics.
- Technologies: **Python**, **PySpark**, **SQL**, **Databricks**, **Apache Spark**, Azure Service Bus, Jinja, Fivetran, Witboost, **Terraform**, **Data Mesh**, Git.

**Junior Software Developer**, Cefriel – Milan, Italy — Feb 2025 – Sept 2025, 8 months

Backend software engineering for M4ESTRO, a Horizon Europe research project focused on resilient manufacturing ecosystems and intelligent supply chains.

- Designed graph-based models and machine learning pipelines for delay prediction and resilience analysis in industrial supply chains.
- Developed cloud-native microservices using **Python**, **AWS** Lambda and **AWS** CDK.
- Contributed throughout the software lifecycle from requirements analysis to deployment within a Horizon Europe research project.
- Technologies: **Python**, **Scikit-Learn**, **AWS** Lambda, **AWS** CDK, **SQL**, **Graph Theory**, **Machine Learning**.

**IT Consultant | Software Engineer**, Andreoli Nastri S.n.c. – Cologno Monzese, Italy — July 2024 – June 2026, 2 years

Designed and developed a complete desktop business application for an Italian manufacturing SME, managing the project from requirements analysis through architecture, implementation and deployment.

- Developed a complete desktop application using **Java**, **Spring Boot** and **Java** Swing, integrating UI, business logic and persistence.
- Designed the relational database model using Spring Data JPA, **Hibernate** and **PostgreSQL**.
- Adopted Neon Serverless **PostgreSQL** and Flyway migrations to provide scalable, low-cost cloud deployment for a small business.
- Technologies: **Java**, **Spring Boot**, **Java** Swing, **Hibernate**, **PostgreSQL**, Neon, Flyway.

## Projects

**AeroFlux** — Nov 2024 – May 2025, 7 months

Cloud-native distributed platform for the safe integration of drones into European airspace. Master's software engineering project focused on distributed systems, autonomous operations and safety-critical software.

- Developed distributed microservices for flight authorization, geo-awareness, weather information and UAV simulation.
- Designed backend services, **REST APIs** and optimized persistence for geospatial workloads.
- Technologies: **Java**, **Spring Boot**, **PostgreSQL**, **MongoDB**, **Docker**, **Kubernetes**.

**Supply Chain Delay Prediction** — Feb 2025 – Sept 2025, 8 months

Master's thesis developed at Cefriel within the Horizon Europe M4ESTRO project. Research combining graph theory and machine learning for predictive analysis of complex supply chain networks. Results are expected to contribute to an Open Research Europe publication.

- Modelled supply chain networks as graphs and developed machine learning models for delay prediction.
- Evaluated graph-based and statistical approaches using industrial data collected within the M4ESTRO Horizon Europe project.
- Technologies: **Python**, **Scikit-Learn**, **SQL**, **Graph Theory**.

**Wavefront Algorithm for Sequence Alignment** — Feb 2023 – July 2023, 6 months

Bachelor's thesis implementing a **Rust** prototype of the Wavefront algorithm for sequence alignment over graph-based genome representations.

- Implemented the Wavefront sequence alignment algorithm in **Rust** for graph-based genome representations.
- Studied optimization strategies for graph algorithms and memory usage.
- Technologies: **Rust**, **Algorithms**, Graph **Algorithms**.

## Skills

**Programming Languages: Java**, **Python**, C++, **Rust**, **SQL**

***Software Engineering*: Java**, **Spring Boot**, **REST APIs**, **Microservices**, **Distributed Systems**, Software Architecture

***Data Engineering*: Python**, **Apache Spark**, **PySpark**, **Databricks**, **SQL**, ETL Pipelines, **Data Mesh**

**Databases: PostgreSQL**, **MongoDB**, **Neo4j**, BigQuery

**Cloud: Docker**, **Kubernetes**, **Terraform**, **AWS**, Git, Linux

**Computer Science: Algorithms**, **Data Structures**, **Graph Theory**, **Machine Learning**

## Languages

**Italian:** Native

**English:** Professional working proficiency (B2)